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ORIGINAL ARTICLE

Evaluating medical student perceptions of family planning and lifestyle during general surgery training

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Abstract

Purpose Medical students' (MS) interest in pursuing general surgery (GS) training is declining, and the effect of perceived lifestyle challenges have on this phenomenon is unknown. This study evaluated MS perceptions of life events, lifestyle, and family planning during GS training.

Methods A voluntary, anonymous survey was distributed to all U.S. medical schools. Logistic regression models were constructed to quantify factors associated with interest in GS.

Results 400 MS from 36 accredited medical schools completed the survey. Most respondents were 21–27 years old (88%), white (58%), female (73%), pre-clinical (65%), and answered that family planning influences their specialty choice (72%). A minority (27%) expressed interest in pursuing GS training. Higher indicated importance of family planning (OR: 0.61, 95% CI 0.38–0.97), lifestyle (OR: 0.14, 95% CI 0.09–0.24), and length of training (OR 0.32, 95% CI 0.19–0.52) were associated with lower interest in GS. Perceptions of gender bias, an inadequate work–life balance, and low level of support for life events in GS training were identified. On multivariable analysis, Black or Other race and perception of support for life events were associated with interest in GS.

Conclusions Overall, MS hold a negative perception of work–life balance and family planning during GS residency. Family planning is valued by MS and associated with a lower likelihood of pursuing GS training. A national commitment is needed to address perceived deficits of support for major life events to increase recruitment of a competitive, diverse workforce.

Keywords General surgery residency · Medical students · Work–life balance · Residency programs · General surgery match

Introduction

The interest of U.S.-trained medical students' (MS) in general surgery (GS) is steadily declining. From 2011 to 2024, an annual average of 5.7% of senior MS from U.S. allopathic medical schools matched into GS; a notable decrease from the average of 6.2% maintained during the 14 years prior [1]. In fact, 2008 was the last year that the percentage of U.S. M.D. seniors matching into general surgery was above 6% [1]. A 20-year retrospective analysis identified a 13% drop in U.S. allopathic seniors matching into GS from 1994 to 2014 [2]. In addition, trends in the National Resident Match Program (NRMP) showed 143 vacant categorical surgical positions [3], while preliminary positions in surgery were filled at less than 65% every year from 2019 to 2024 [1]. Given the startling decrease in general surgery matriculants over the last decade, there is a concern for a substantial shortage in surgeons as soon as 2025 [4].

Multiple factors are thought to influence MS interest in GS. Interest in a surgical specialty is high during the preclinical years [5, 6] and some studies suggest that students are attracted to the field because of unique intellectual challenges and ability to perform procedures [5, 7]. It is difficult

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to determine what plays the most important role in career selection, but factors, such as mentorship, timing of exposure, and clerkship experience, have previously been identified as important factors in this decision [8–12].

General surgery as a specialty faces difficulties in recruiting students due to negative perceptions of the career's lifestyle. 70% of MS do not believe general surgeons live well-balanced lives [13], and despite implementation of 80-h work week policies, early, pre-clinical interest in the field has not correlated with an increase in matriculation [14]. Lifestyle is important to MS when selecting careers [15]; this includes being able to start families during residency if desired [13]. However, within GS, there is little consensus from program directors regarding nationwide policies for parental leave, child care, and lactation spaces during residency [16, 17]. Minimal research has been done to reassess how students perceive opportunities to initiate family planning during surgical residency. Thus, the goal of this study was to evaluate current MS perceptions toward family planning and parenthood during GS residency. Second, we aimed to identify potential factors that hinder MS matriculation into general surgery.

Methods

Survey administration

An anonymous, electronic survey was distributed to MS Surgery Interest Group leaders with contact information provided via the American College of Surgeons webpage [18]. These MS leaders and personal MS contacts from our institution's dedicated education research collaborative were encouraged to respond and to distribute the survey on public platforms connected to their respective medical schools (e.g., group chat, email listserv, MS social media page, etc.). The decision to utilize personal MS contacts allowed for a familiar point of contact for survey distribution within each respective medical school and better ensured closed loop communication to our research team regarding successful survey deployment. This also allowed for an individualized approach to survey distribution to capitalize on the most frequently visited platforms at each school. Study data were collected and managed using REDCap electronic data capture tools. Inclusion criteria consisted of MS currently enrolled in any Liaison Committee on Medical Education (LCME) or Commission on Osteopathic College Accreditation (COCA) accredited M.D., M.D.-Ph.D., or D.O. program in the US between January 2022 and March 2022. All contacts were provided with an introductory letter that included the study goals and a unique survey link. Institutional contact points then chose their preferred method of survey delivery, which included distribution via email or Class GroupMe pages. Responses were voluntary, anonymous, and consent was inferred by each participant choosing to complete the survey.

Survey design

This survey was composed of 27 questions and was organized into four sections (Supplementary Materials). These sections included demographic information, level of interest in GS, perceptions of family planning during GS training, and open-ended commentary. Question format consisted primarily of a 5-point Likert scale, multiple-choice, yes/no, and free responses. This survey was developed using a modified Delphi method to achieve group consensus to format the questionnaire. Further, this was facilitated through a dedicated education research collaborative of surgical faculty, surgical residents, statisticians, and medical students at a single institution.

Statistical analysis

Baseline demographic characteristics of all respondents were evaluated. Medical students were grouped into cohorts based upon expressed interest in pursuing GS training (Yes vs. No). Demographic characteristics and survey responses were compared between cohorts. Medical students were further stratified by medical education level (pre-clinical vs. clinical) for subset comparison of survey responses. Univariable and multivariable logistic regressions were performed to identify predictors of interest in pursuing a GS residency.

Categorical data were reported as total (n) and percentage (%) and analyzed using chi-square or Fischer's exact tests as appropriate. Continuous data were reported as median and interquartile range (IQR) and analyzed using Wilcoxon rank-sum test. Likert scale scores were grouped into binary yes/no answers by combining high negative (1), low negative (2) and neutral (3) scores into "no/negative" and low positive (4) and high positive (5) scores into "yes/positive".

On univariable and multivariable logistic regression, odds ratios (OR) were reported with 95% confidence intervals. To account for confounding, multivariable logistic regression was performed using predictors found significant with a $p < 0.2$ on univariable analysis. The multivariable model was then reduced in a backwards stepwise approach. A p value of < 0.05 considered statistically significant. Statistical analysis was performed via JMP Pro version 16.0 (Cary, NC: SAS institute). This study was approved by the Institutional Review Board (IRB#2021-0465).

Results

A total of 400 participants from 36 medical schools completed the survey. Demographic information is summarized in Table 1. Most respondents (88%) were between

the ages of 21 and 27 and a majority (74%) identified as female. 58% of respondents identified as White race while 17% as Black, 13% as Asian, 3% as Hispanic/Latino, and 10% as Other (American Indian or Alaska Native, Middle Eastern or Northern African, Native Hawaiian or Other Pacific Islander). Most respondents were from institutions in the Southern (49%) and Midwestern (45%) regions of the United States. 65% were in their pre-clinical years of medical school. 65% of respondents reported being in a relationship. Most (95%) did not have children although 72% of respondents reported that family planning influences their choice of residency. Notably, only 27% of respondents (28% pre-clinical respondents and 24% clinical) were interested in pursuing a GS residency for their career.

Figure 1 represents MS perceptions of family planning, parenthood, and gender-specific experience in GS. 84% of respondents agreed that GS residents delay starting families during training due to the rigor and time commitment. 91% did not believe GS residents Are readily able to start families. 86% of respondents reported that GS residents Are not supported by their colleagues once having children. 95% of respondents did not believe that GS residents have proper parental support (i.e., onsite child care, lactation facilities, or adjusted schedules) at hospitals. In addition, 96% did not believe that GS residents maintain a healthy work–life balance, and 97% did not believe that GS residents have adequate time for parental leave. Overall, 99% reported that GS residents did not have enough time to spend with family and children during training. When asked about gender-specific residency experiences, 74% of respondents did not believe GS is suited to all gender identities and 84% did not believe MS of all gender identities are treated equally during the GS rotation. The majority of MS (97%) responded with interest in more than one specialty, including 25% of MS indicating interest in GS and another specialty.

Table 1 Demographic characteristics of medical student survey respondents

Characteristics ($N=400$)	N (%)
Age	
21–27	352 (88.0)
28–34	39 (9.8)
35–41	7 (1.8)
42–48	2 (0.5)
Gender identity (female)	293 (73.5)
Race or ethnicity	
Asian	52 (13.1)
Black or African American	69 (17.4)
Hispanic/Latino	10 (2.5)
White	228 (57.6)
Other	6 (1.5)
Multiracial	31 (7.8)
Region	
Midwest	180 (45.1)
South	194 (48.6)
Northeast	21 (5.3)
West	4 (1.0)
Medical school level [pre-clinical (MS1, MS2)]	257 (64.7)
Relationship status (single)	140 (35.0)
Do you have children? (yes)	22 (5.5)
Does family planning influence choice of medical specialty? (yes)	287 (71.9)
Are you interested in general surgery? (yes)	106 (26.5)

MS demographics relative to interest in GS are reported in Table 2. Age, gender, medical education level, intended degree, and relationship status were not significantly different among MS who are interested in GS in comparison to those who did not express interest. A higher proportion of MS identifying as Black, Asian, or other race and a lower proportion of White or Hispanic expressed overall interest in GS ($p<0.01$). Interest in GS among MS was associated with perceptions of work–life balance, gender, and family planning (Fig. 2) (Supplemental Table). There were no significant differences in MS perceptions of family planning, time spent with family, the impact of childbearing on surgical education, or parental support during GS residency based on interest in GS. However, MS with reported interest in GS were more likely to believe GS residents are supported by their colleagues when choosing to have children during training ($p<0.01$) and have a healthy work–life balance ($p=0.02$). Additionally, MS interested in GS were more likely to believe the specialty is equally suited to all gender identities ($p<0.01$).

A subset analysis comparing the perceptions of pre-clinical to clinical medical students regarding GS training is shown in Table 3. Compared to pre-clinical MS, clinical MS were more likely to agree that they encounter general surgeons of the same gender identity as themselves (OR: 2.30, 95% CI 1.52–3.52; $p<0.01$) and endorse that MS of all gender identities are treated equally on GS rotations (OR: 1.92, 95% CI 1.12–3.31; $p=0.02$). All other GS residency lifestyle survey responses were similar between pre-clinical and clinical MS.

On sensitivity analysis of survey responses by medical education level and interest in pursuing GS residency, pre-clinical MS who expressed interest in GS and clinical MS who were not interested were more likely to agree that they encounter general surgeons of the same gender identity as themselves, compared to pre-clinical, uninterested MS. Clinical MS who expressed interest in GS training had equal odds to pre-clinical, uninterested MS of endorsing encountering general surgeons of the same gender identity as themselves. Additionally, clinical MS who were interested and clinical MS who were

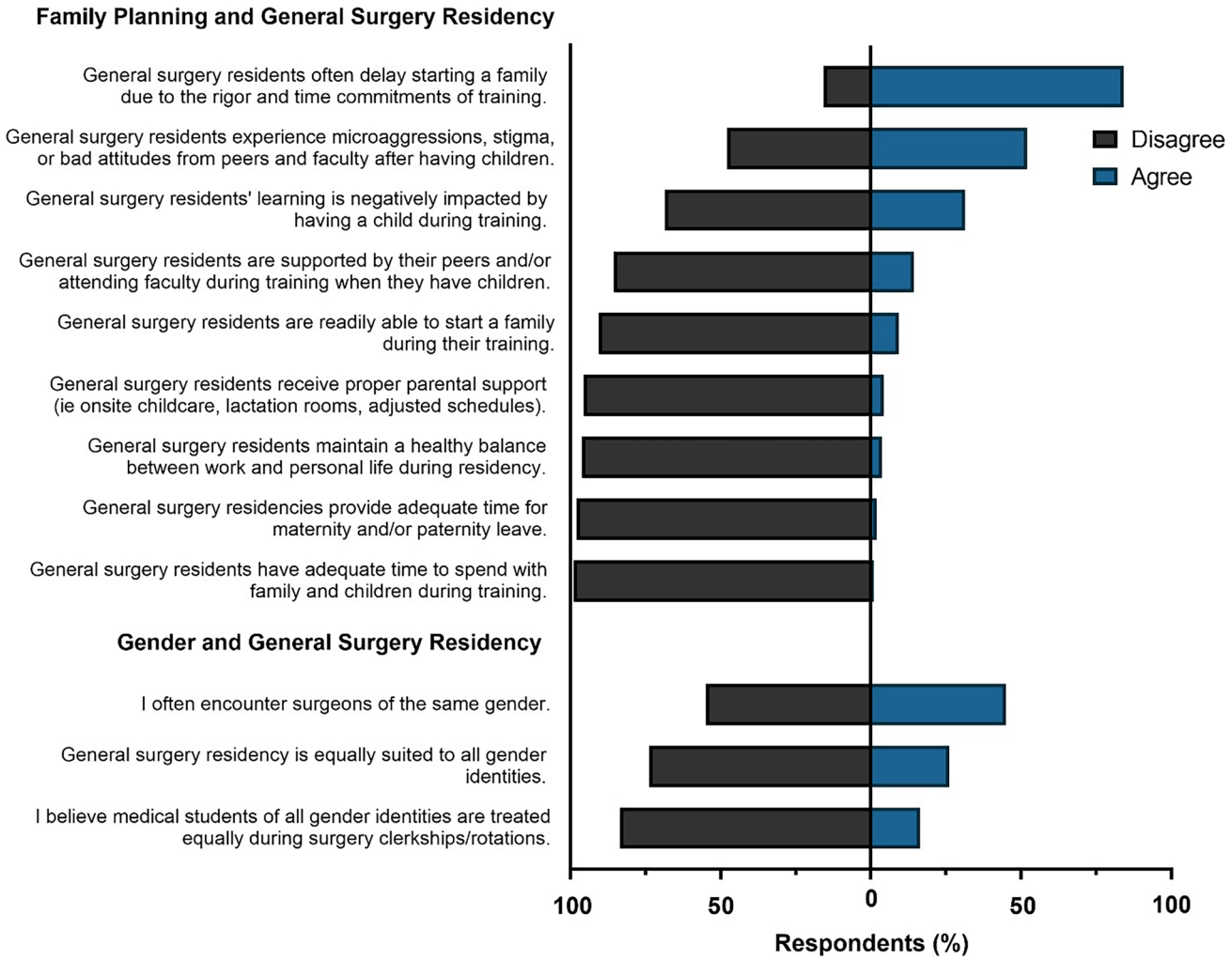


Fig. 1 Responses to survey questions regarding medical students' perceptions of gender bias, work–life balance, and family planning in general surgery residency training

uninterested in GS residency were more likely to identify that MS of all gender identities were treated equally compared to pre-clinical, uninterested MS.

Multivariable logistic regression analysis of predictors of interest in pursuing a GS residency for MS is shown in Table 4. A significant positive association exists between the respondents who identified as Black or Other race with an interest in pursuing a GS residency ($p < 0.01$ for both). In addition, the perception of support for residents with children from colleagues during residency was positively associated with MS interest in pursuing a GS residency ($p = 0.02$). However, the perception of equal treatment of MS on GS rotation, that GS residency is suited to all genders, and that GS residents have a healthy work–life balance were not associated with interest in GS.

Discussion

This is cross-sectional study was performed to assess MS perceptions of family planning, parenthood, and lifestyle during GS residency, as well as to identify factors that influence their interest in the field. Contributing to a body of literature that was published a decade or two previously with few interim analyses or updates, we found that MS hold overwhelmingly negative perceptions of these experiences. Furthermore, it was observed that certain race/ethnicities and beliefs regarding family planning may be predictive of GS interest. Approximately 40–60% of MS are interested in surgery at the start of medical school or even before entering matriculation, but change their minds

Table 2 Medical student survey respondent demographics by interest in general surgery

Variables	Interested in general surgery $N=106$ (26.5%)	Not interested in general surgery $N=294$ (73.5%)	p value
Age			0.52
21–27	90 (84.9)	262 (89.1)	
28–34	13 (12.3)	26 (8.8)	
35–48	3 (2.8)	6 (2.0)	
Race/ethnicity			**<0.01**
White	45 (42.5)	183 (62.2)	
Asian	15 (14.2)	37 (12.6)	
Black	29 (27.4)	40 (13.6)	
Hispanic	1 (0.9)	9 (3.1)	
Other	16 (15.1)	25 (8.5)	
Gender (female)	79 (74.5)	214 (73.0)	0.77
Medical education level (pre-clinical)	33 (31.4)	107 (36.6)	0.34
Year of graduation			0.33
2022	6 (5.7)	37 (12.8)	
2023	18 (17.0)	50 (17.2)	
2024	33 (31.1)	84 (29.0)	
2025	40 (37.7)	115 (39.7)	
2026–2028	3 (2.8)	4 (1.4)	
Degree expected (M.D.)	95 (89.6)	264 (90.7)	0.74
Region			0.37
Midwest	43 (42.6)	137 (46.6)	
Northeast	8 (7.9)	13 (4.4)	
South	50 (49.5)	144 (49.0)	
Relationship status (single)	44 (41.5)	96 (32.7)	0.10
Partner employed (yes)	45 (72.6)	132 (66.7)	0.38
Hours per week partner works			0.97
<40	10 (22.2)	27 (20.5)	
40–65	32 (71.1)	96 (72.7)	
>65	3 (6.7)	9 (6.8)	
Partner is a medical student, resident, or attending (yes)	17 (27.4)	49 (24.8)	0.67
Children (yes)	4 (3.8)	18 (6.1)	0.37

Bold indicates statistical significant with p<0.05

as years progress [6, 19]. Our results are congruent with prior older studies that demonstrated that students are deterred from pursuing general surgery due to factors like a demanding lifestyle and long work hours [11, 20–22].

Previous studies have investigated factors influencing MS career decisions in more detail. Hill et al. studied a cohort of pre-clinical and clinical students and found that MS often hold negative opinions of surgeons and surgery, which can adversely influence their clinical experience and dissuade them from pursuing a career in surgery [23]. Previously, Sanfey et al. found that 70% of MS believed that surgeons did not have well-balanced lives, which served as a major obstacle to entering the field [13]. Our study verifies these findings, as we observed that most respondents did not believe that surgical residents were able to readily start families, spend adequate time with their families, have a healthy work–life balance, or even take sufficient parental leave. These negative perceptions Are particularly worrisome considering they persist two decades following the establishment of the 80-h resident work week which, along with other national policies, was intended to protect off-duty hours for trainees. Plainly, increased interest in general surgery residency programs has not been seen following the mandated safeguards for resident well-being, with the annual enrollment of U.S. seniors remaining below that of pre-policy rates [14, 24].

Our findings may reflect overarching perceptions that work–life balance is not attainable during GS residency, which poses significant issues to recruitment and retention and should be a call for change in the training paradigm.

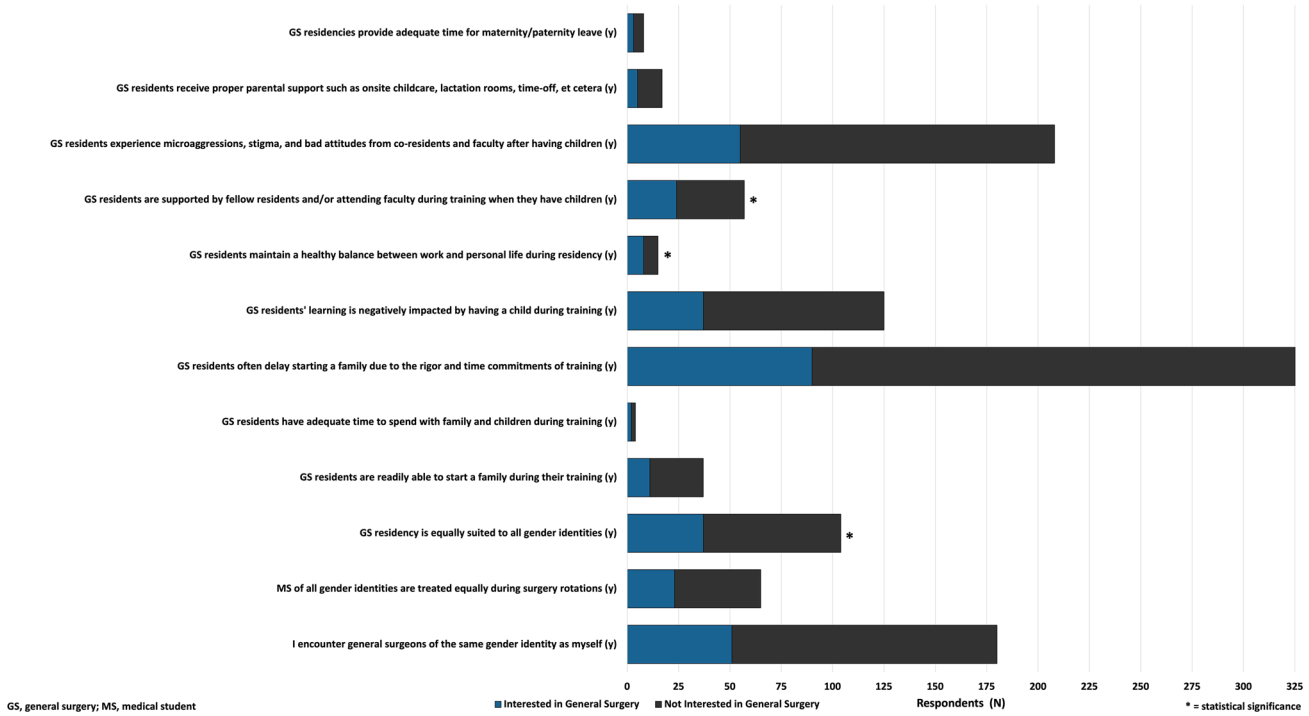


Fig. 2 Responses to survey questions regarding medical students' perceptions of gender bias, work–life balance, and family planning in general surgery residency training by indicated interest in general surgery versus no interest

Table 3 Subset comparison of medical students' perceptions of gender bias, work–life balance, and family planning categorized by medical education level and interest in pursuing a general surgery residency

Variables	Odds ratio	95% CI		*p* value
I encounter general surgeons of the same gender identity as myself (y)				
Pre-clinical	Ref			
Clinical	2.30	1.52	3.52	**<0.01**
Medical students of all gender identities are treated equally (y)				
Pre-clinical	Ref			
Clinical	1.92	1.12	3.31	**0.02**
I encounter general surgeons of the same gender identity as myself (y)				
Pre-clinical not interested	Ref			
Pre-clinical interested	1.78	1.02	3.09	**0.04**
Clinical not interested	3.07	1.88	5.06	**<0.01**
Clinical interested	1.87	0.88	3.96	0.10
Medical students of all gender identities are treated equally (y)				
Pre-clinical not interested	Ref			
Pre-clinical interested	1.81	0.83	3.82	0.13
Clinical not interested	2.12	1.10	4.13	**0.03**
Clinical interested	3.08	1.21	7.41	**0.02**

Bold indicates statistical significant with p<0.05

We found that, regardless of gender, applicant demographic, or interest in the field, MS prioritize the ability to start families when pursuing a surgical residency and hold negative beliefs about the experience in the context of GS training. In a single institutional study, Giantini Larsen et al. evaluated potential obstacles to pursuing GS and found no significant differences between male and female MS with respect to their inclination toward the field [25]. Nonetheless, their investigation revealed that female respondents expressed heightened apprehension regarding the impact of GS training on their dating prospects, family commitments, ability to bear children, and access to maternity leave. Male respondents were also influenced by factors related to family planning, which prompted them to consider alternative career paths [25]. Although gender differences in perceptions of family planning were not apparent in our study, our findings imply that MS, irrespective of gender, regard these experiences as critical to achieving work–life balance during their training.

The negative perceptions regarding GS residency Are significant, and it is worth considering the causes of and validity of student concerns. In previous evaluation, approximately 30% of female surgical residents do not feel supported by colleagues or senior faculty when choosing to take parental leave, and very few could identify current Accreditation Council for Graduate Medical Education (ACGME) guidelines on parental leave [26]. Duty hour restrictions

Table 4 Multivariable analysis of predictors of interest in pursuing a general surgery residency

Variables	Odds ratio	95% CI		*p* value
Race/Ethnicity				
White	Ref			
Asian	1.70	0.84	3.42	0.14
Black	3.22	1.74	5.92	**<0.01**
Hispanic	0.40	0.05	3.42	0.41
Other	2.77	1.32	5.78	**<0.01**
Medical students of all gender identities are treated equally (y)	1.45	0.77	2.72	0.25
GS residency is equally suited to all gender identities (y)	1.69	0.99	2.89	0.06
GS residents maintain a healthy balance between work and personal life during residency (y)	2.03	0.67	6.20	0.21
GS residents are supported by fellow residents and/or attending faculty during training when they have children (y)	2.03	1.08	3.82	**0.02**

Bold indicates statistical significant with p<0.05

show varying effects on surgical resident's well-being and quality-of-life, with some studies showing no improvement to wellness and even a poor understanding of universal policies for parental leave [26, 27]. Therefore, it is uncertain whether MS are aware of current parental policies and how they might benefit long-term family goals. Future studies should aim to assess MS understanding of duty hour restrictions and investigate resident experiences with family planning, specifically in the context of these negative perceptions.

It is encouraging to discover that MS who are interested in general surgery tend to believe that residents are supported by both their colleagues and attending faculty when it comes to parenting. These findings may imply that MS interested in general surgery hold more optimistic views regarding the current culture and factors that influence well-being of trainees, despite some existing uncertainties. Future work could evaluate differences in viewpoints GS residency training among MS who express interest in both surgical and non-surgical fields. This may help elucidate critical career decision branch points or reinforce the MS concerns we identified.

In addition, it is important to note that most of our respondents were in their pre-clinical years and had not yet completed a GS rotation. Cochran et al. investigated MS' perceptions of GS both before and after completion of the GS clerkship and found that their perceptions of general surgery improved, but their level of interest in surgery remained unaffected [8]. Our study similarly found that clinical experience was associated with improved perceptions in treatment of MS during GS clerkships. Even so, MS opinions regarding GS residency lifestyle challenges were no different between pre-clinical and clinical MS regardless of expressed interest in the field.

Cook et al. found that MS who participated in rotations with more personalized mentorship and teaching were more likely to be interested in comparison to those who completed a standard rotation at an academic center [28], highlighting a potential area where departments can increase interest. More recently, Patterson et al. reported that mentorship was a critical component when considering a surgical career [12]. Intentional involvement in surgical interest groups, shadowing programs, and mentorship programs that highlight different careers in surgery can be advantageous in recruiting prospective surgeons, increasing interest, and eliminating negative opinions concerning family life and surgical training [29–31].

There Are several limitations to our study. We had a relatively small sample size with only 400 respondents, which may limit the breadth of opinions captured in the study. There was also a gender imbalance in survey respondents, with a higher portion of female participants. This imbalance presents a challenge in truly discerning if there are gender-based differences in perceptions of GS. In addition, respondents were primarily from the Midwest and South, which may introduce regional bias in our results. Respondents may not have answered truthfully if the security anonymity was of concern. And while the survey questions were carefully selected, worded, and workshopped and study aims were provided in conjunction with survey distribution, respondents may have misunderstood the intent of the questions or survey. Additionally, most respondents indicated interest in multiple fields, rather than GS alone, so a subset analysis of isolated GS interest compared to mix or non-GS interests could not be completed. Finally, as there was no comparison group regarding perceptions of an alternative residency, respondents may have been biased toward responding to questions about general surgery and/or family planning and lifestyle challenges. These responses may not be representative of the true perceptions of all medical students. To truly identify variations in MS' perceptions of general surgery, a broader sample of students should be recruited from across

the US. Future studies should attempt to evaluate the importance of other factors, such as mentorship and pre-clinical activities, in MS' interest in GS.

Conclusions

Medical students have negative perceptions surrounding the experience of family planning, parenthood, and lifestyle in GS, which may be a significant deterrent to pursuing this career path. However, students who are interested in GS hold some positive perceptions regarding elements of parental experience and support during training. To fully address this, future studies should explore ways to expand MS exposure to GS and provide insight into the experiences of residents and faculty members. It is crucial to investigate the current experience of family planning during residency to elucidate challenges and implement targeted policies to improve the experience of future trainees.

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Data availability The authors confirm that the data supporting the findings of this study are available within the article. The raw data that support the findings of this study are available from the corresponding author, upon reasonable request.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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